Computer and Network Security (2026-2027)

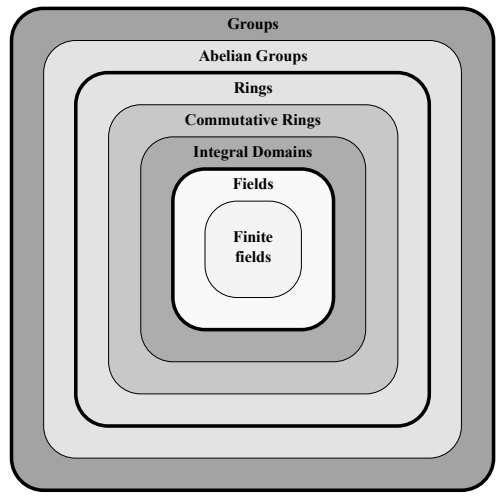
Part 2: Homework – Finite Fields

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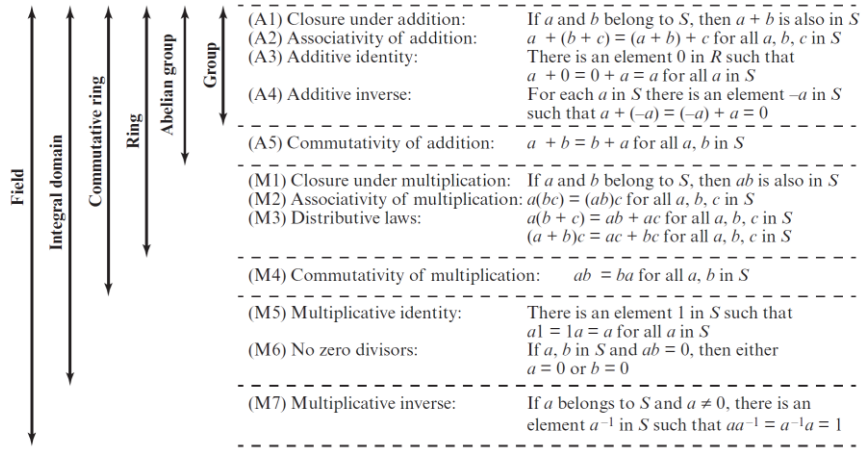
Groups, rings, and fields



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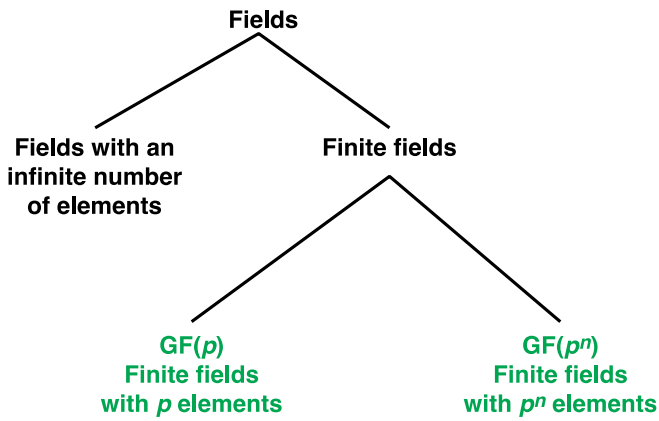
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Properties



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Finite fields



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The GF(2^n) field: $f(x) = \sum_{i=0}^{n-1} a_i x^i$

- Each a_i is a binary value, resulting in 2^n polynomials
  - e.g., for n = 3: 0, x, x^2, $x^2 + x$, 1, $x + 1$, $x^2 + 1$, and $x^2 + x + 1$
- Arithmetic operation definitions
  - The basic polynomial arithmetic rules are followed, except:
  - Arithmetic on the coefficients is performed modulo 2 (i.e., XOR)
  - If the result of multiplication has a degree larger than x^{n-1} then reduce it modulo some irreducible polynomial* m(x)
- Two possibilities for m(x) when n = 3:
  - $x^3 + x^2 + 1$
  - $x^3 + x + 1$

*A polynomial f(x) over a field F is called **irreducible** if and only if f(x) cannot be expressed as a product of two polynomials, both over F, and both of degree lower than that of f(x).

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Finite fields in AES

- AES requires a set of integers that:
  - Support division (i.e., each nonzero element has a multiplicative inverse)
  - All fit in n bits as to not waste space (i.e., a range from 0 to 2^n-1)
- The set of integers $\mathbb{Z}_{2^n}$ does not satisfy these requirements:
  - e.g., 2 does not have a multiplicative inverse, i.e., there is no b such that $2b \ mod \ 2^n = 1$
- A finite field of type GF(2^n) satisfies these requirements
  - It consists of 2^n elements, each represented by a polynomial f(x):

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0 = \sum_{i=0}^{n-1} a_i x^i$$

  - Where each a_i is a binary value (i.e., 0 or 1), and f(x) thus represents a binary number < 2^n
- **AES uses GF(2^8) with m(x) = $x^8 + x^4 + x^3 + x + 1$**

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Addition in GF(2^8)

Addition is performed as a bit-per-bit XOR

Example

$$\begin{array}{rl} x^7 + x^5 + x^4 + x^3 + x & (= 1011\ 1010 = 0xBA) \\ + \quad x^3 + x + 1 & (= 0000\ 1011 = 0x0B) \\ \hline x^7 + x^5 + x^4 + 1 & (= 1011\ 0001 = 0xB1) \end{array}$$

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Multiplication in GF(2^8)

Multiplication can be performed as polynomial multiplication modulo m(x)

Example

$$\begin{array}{r} x^7 + x^5 + x^4 + x^3 + x \\ \times \quad x^3 + x + 1 \\ \hline x^7 + x^5 + x^4 + x^3 + x \\ x^8 + x^6 + x^5 + x^4 + x^2 \\ + \ x^{10} + x^8 + x^7 + x^6 + x^4 \\ \hline x^{10} + x^4 + x^3 + x^2 + x \end{array}$$

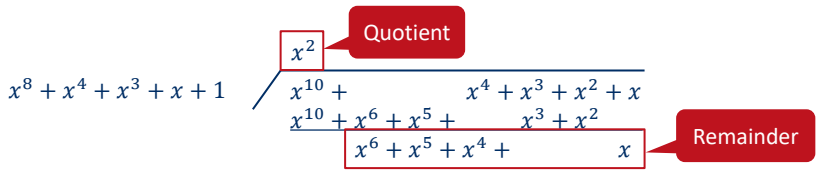
Problem: The solution c(x) does not fall within GF(2^8)
Solution: calculate c(x) mod $x^8 + x^4 + x^3 + x + 1$

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Multiplication in GF(2^8)

c(x) mod m(x) can be found by calculated the remainder of c(x) / m(x)



Result

$a(x) = x^7 + x^5 + x^4 + x^3 + x, \quad b(x) = x^3 + x + 1$

$a(x) \times b(x) = x^6 + x^5 + x^4 + x = 0111\ 0010 = \{72\}$

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Multiplicative inverse in GF(2^8)

- The multiplicative inverse of a(x) is $a^{-1}(x)$ if: $a(x) \times a^{-1}(x) \equiv 1 \pmod{m(x)}$
- The value of $a^{-1}(x)$ can be found by using the **Extended Euclidian Algorithm**:

Initialization

$r_{-1}(x) = m(x)$ $\quad w_{-1}(x) = 0$

$r_0(x) = a(x)$ $\quad w_0(x) = 1$

Recursive algorithm

$r_i(x) = r_{i-2}(x) \bmod r_{i-1}(x)$

$q_i(x) = r_{i-2}(x) / r_{i-1}(x)$

$w_i(x) = w_{i-2}(x) - q_i(x)\ w_{i-1}(x)$

Stop condition

if $r_i(x) = 1$ then $w_i(x) = a^{-1}(x)$

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Multiplicative inverse in GF(2^8)

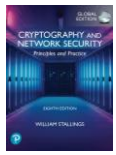
Example

Calculate $a^{-1}(x)$, given $a(x) = 0x22 = 0010\ 0010 = x^5 + x$

Solution (using Extended Euclidian Algorithm):

1. $r_1 = x^3 + x + 1, q_1 = x^3, w_1 = x^3$
2. $r_2 = x^2 + 1, q_2 = x^2 + 1, w_2 = x^5 + x^3 + 1$
3. $r_3 = 1, q_3 = x, w_3 = x^6 + x^4 + x^3 + x = 0101\ 1010 = \{5A\} = a^{-1}(x)$

Further reading on finite fields



- Chapter 5 (Section 5.1—5.6)